Jayanth Vallabi

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SUMMARY

Results-driven Python Developer with 4+ years of experience building scalable backend and full-stack applications using Python, Django, Flask, FastAPI, React, and REST APIs. Experienced in AWS, Docker, Kubernetes, microservices, database development, automation, and Agile delivery, with a strong focus on performance, reliability, and business impact.

SKILLS

- **Programming Languages:** Python (3.x), JavaScript, R, Java, C, Erlang, Prolog
- **Python Libraries:** Pandas, NumPy, SciPy, scikit-learn
- **Web & API Development:** Django, Flask, FastAPI, RESTful APIs, GraphQL, React.js, AngularJS, HTML5, CSS3, Bootstrap
- **Databases:** MySQL, PostgreSQL, Microsoft SQL Server, SQLite, MongoDB, Oracle, ChromaDB
- **Cloud Platforms:** AWS (EC2, S3, RDS), Microsoft Azure, Google Cloud Platform (GCP)
- **DevOps & Infrastructure:** Docker, Kubernetes, Jenkins, Terraform, Ansible
- **Version Control & Collaboration:** Git, Bitbucket, JIRA, Confluence
- **Data Visualization & Tools:** Tableau, HPQC, FuelTrak
- **IDEs & Development Tools:** VS Code, PyCharm, IntelliJ IDEA, Eclipse, Atom, Spyder
- **Methodologies:** Agile, Scrum, Test-Driven Development (TDD)

EXPERIENCE

Software Development Engineer

Cigna Healthcare Inc. **January 2025 - Present, USA**

- Developed and maintained a secure, HIPAA-compliant patient data ingestion system using **Python, Django REST Framework, and FHIR standards**, enabling real-time EHR access and reducing data retrieval latency by **30%**.
- Contributed to the implementation of a **microservices-based architecture on AWS** using **S3, Lambda, EC2, and API Gateway**, supporting **99.9% uptime** and scalable access across **4 departments**.
- Automated healthcare claims validation workflows using **Pandas, Scikit-learn, and business rules**, achieving **96% accuracy** and reducing manual effort by more than **115 hours per month**.
- Built a full-stack predictive analytics dashboard using **Python, Flask, React, and Tableau**, collaborating with data scientists and QA teams to analyze treatment outcomes and support a **15% reduction in patient readmission rates**.
- Supported the migration of over **5 million medical records** from legacy systems to **PostgreSQL**, ensuring **zero data loss** and regulatory compliance through data validation, **QA** testing, and **Pytest** automation.

Python Developer

Capgemini **May 2021 - December 2023, India**

- Developed and maintained backend services using **Python, Django, and REST APIs** to integrate patient health records with healthcare platforms, supporting reliable data synchronization across client-facing applications.
- Contributed to a scalable **microservices architecture** using **Docker and Kubernetes**, improving system efficiency by **20%** and reducing application downtime by **30%**.
- Collaborated with front-end teams to develop responsive user interfaces using **React, Angular, HTML5, and CSS3**, helping reduce UI-related bug tickets by **30%**.
- Participated in **Agile sprint planning, code reviews, and cross-functional meetings**, using **JIRA, Confluence, and Bitbucket** to support documentation, version control, and collaborative development workflows.

EDUCATION

Master of Science in Computer Science

California State University Fullerton • USA • 2026

PROJECT

Healthcare Document Intelligence Platform

- Built a **Python-based document intelligence platform** using **FastAPI**, **LangChain**, **Hugging Face** Transformers, **RAG**, and **Pinecone** vector database, enabling semantic search and **Q&A** over **10,000+** clinical documents with retrieval response time under **2** seconds.
- Developed **REST APIs** and microservices using **FastAPI**, **Pydantic**, **SQLAlchemy**, and **PostgreSQL** with **JWT** authentication, **asynchronous** programming, and OOP design patterns, supporting **500+** concurrent API requests with automated pytest test coverage.
- Used **R** for data analysis and validation of clinical document datasets to support evaluation of document processing and retrieval results.
- Deployed the application as a **containerized microservice** using Docker, Kubernetes, AWS Lambda, and CI/CD with GitHub Actions, achieving **99%** uptime and **zero-downtime deployments** with automated monitoring.